ECOLOOP — TECHNICAL & CAPABILITIES DOSSIER

Complete System Capabilities, Tech Stack Architecture & Hackathon Technical Q&A Guide

System Core Purpose: EcoLoop is a decentralized, full-stack campus circular economy platform that digitizes e-waste collection, eliminates the first-mile disposal barrier for students, enforces tamper-evident QR chain of custody, and generates real-time statutory environmental audits.

SECTION 1: WHAT ALL THE WEBAPP CAN DO (10 CORE CAPABILITIES)

01 Master Recovery Catalog (55+ Verified Campus Devices)

Categorized database covering personal student gear (laptops, chargers, wearables) and departmental lab clusters (servers, oscilloscopes). Instantly calculates standard average mass, recoverable precious metals, and toxic hazards under CPCB 2022 rules.

02 Digital Hardware Passport & Dynamic QR Tagging

Every registered item generates an immutable digital identity token with an exportable, printable QR code. Eliminates untracked "black-box" e-waste bins by establishing origin identity down to the specific student, hostel, or lab room.

03 In-Browser Optical QR Scanner (Zero App Installation)

Uses HTML5 video stream and WebAssembly canvas decoding to scan physical QR codes directly on any Android, iOS, or laptop browser. Campus EcoSquad volunteers verify drop-offs on the spot without installing APKs.

04 Campus Operations Hub & Multi-Stage Pipeline

Central control room managing all active requests across 5 tamper-evident states: [Registered -> Scheduled -> Collected -> Handed Over -> Verified Recycled]. Logs timestamps, handler IDs, weights, and transit GPS points.

05 Calibrated Scale Weigh-in & Instant Point Attribution

Prevents cheating and contaminated waste by recording scale-verified mass during physical handover. Automatically computes reward points (10 pts/kg) credited live to student accounts with animated UI telemetry.

06 Certified Recycler Ingestion & Dismantling Certificates

Specialized interface for authorized CPCB recyclers to ingest campus batches, record verified pyrometallurgical yields, and issue tamper-evident digital green recycling certificates back to the student.

07 Scope-3 Carbon Offset Engine (GEM 2024 Standard)

Calculates verified avoided emissions using empirical 1.44 kg CO2e / kg factor. Equates savings to tree plantings and smartphone recharges, generating downloadable signed PDF accreditation records.

08 Statutory 30-Day CPCB Compliance SLA Engine

Enforces Rule 13 of India E-Waste Management Rules 2022, which mandates campus storage cannot exceed 30 days. Features real-time countdown clocks, threshold warnings, and automated breach alerts to administrators.

09 One-Click Official NAAC Criterion 7.1.3 Audit Dossier

Automates university environmental inspection reporting. Compiles gross mass diverted, authorized recycler chain-of-custody receipts, and hostel participation breakdown into a printable official NAAC dossier.

10 Municipal Portal & CPCB City Hotspot Vector Network

Interactive spatial telemetry for civic bodies (MCC, CMC, BBMP) and NGOs. Maps regional e-waste hotspots, shares anonymized campus diversion data, and schedules bulk citizen community collection drives.

SECTION 2: COMPLETE TECHNOLOGY STACK & SYSTEM ARCHITECTURE

TECHNICAL STACK BREAKDOWN (ENTERPRISE / AEROSPACE GRADE)

Frontend Layer:	**Next.js 16.4 (App Router) + React 19.3**	— Server-Side Rendering (SSR) & Partial Prefetching for sub-second page delivery.
Design & UI System:	**Tailwind CSS v4 + Vanilla Tokens**	— Austere SpaceX monochrome aesthetic (#000000, #ffffff, #3a3a3f), zero clutter.
Vision & Optical IO:	**HTML5-QRCode Engine + QRCodegen**	— Hardware-independent optical barcode decoding directly within mobile web browsers.
Backend REST API:	**Node.js 24 + Express 4.21 (ES Modules)**	— Modular routing architecture with comprehensive error handling & CORS middleware.
Data & Persistence:	**MongoDB Atlas + Resilient Memory Store**	— Dual-mode storage architecture: instant auto-seeding with zero cloud dependency.
Security & Auth:	**JWT (HMAC-SHA256) + bcryptjs (10 rounds)**	— Role-based access control (Student, Collector, Admin, Recycler) with strict hashing.
Document Engine:	**PDFKit 0.20 Server-Side Stream**	— Programmatic vector generation for NAAC audits & signed carbon certificates.
API Standards:	**OpenAPI 3.0 / Swagger UI Interactive**	— Public REST endpoints for inter-college federation and municipal integrations.

SYSTEM INTERACTION & DATA FLOW ARCHITECTURE

1. CLIENT TIER
Next.js 16 (App Router)
Mobile Web, PWA UI, Optical QR viewfinder, telemetry displays

2. ROUTING & GATE
Vercel Edge Gateway
Global CDN caching, automatic HTTPS, DNS load balancing

3. BACKEND API
Express REST Service
Auth validation, CPCB SLA checks, carbon calculation engines

4. DATA & SYNC
Dual Persistence Engine
MongoDB Atlas or zero-downtime auto-seeding in-memory store

ARCHITECTURAL INNOVATIONS & DESIGN DECISIONS

1. Resilient Dual-Mode Data Store (Zero-Failure Guarantee)

If external MongoDB Atlas credentials are missing or experiencing network drops, the backend automatically fails over to an internal, in-memory store pre-seeded with official campus ERP records. The application NEVER crashes or presents blank screens during live demonstrations.

2. Hardware-Agnostic WebAssembly Optical Scanning

Rather than requiring students or volunteers to download a native Android/iOS APK from app stores, EcoLoop executes browser-native HTML5 barcode parsing. Camera frames are processed directly in client memory with zero image transfer over the network, ensuring high battery efficiency and low bandwidth usage.

3. Cryptographic DPDP Act 2023 Compliance

Complies with India's Digital Personal Data Protection Act. When campus e-waste data is aggregated for municipal partners or research papers, student names, phone numbers, and hostel room numbers are stripped using a one-way anonymization protocol.

4. Dynamic Client Fallback & Vercel Serverless Integration

The client dynamically evaluates window location at runtime: running against localhost in development, and routing seamlessly to high-availability serverless microservices in production without manual environment rebuilding.

SECTION 3: TECHNICAL QUESTIONS & ANSWERS (JUDGES CHEAT SHEET)

Q1: How does your optical QR scanner work without a native mobile app?

We leverage the modern WebRTC MediaDevices API (getUserMedia) combined with an optimized JavaScript/WebAssembly barcode decoding worker (html5-qrcode). The video stream is processed entirely within a browser canvas on client side. It runs on any mobile device (Android Chrome, iOS Safari) with zero installation friction.

Q2: How do you prevent students from cheating the system or entering fake weights?

Self-registration in the portal only logs an estimated baseline. Point attribution and status transition to [collected] are physically locked to the campus EcoSquad collector role. The collector must physically scan the item's QR code, inspect the hardware category, and input the verified scale reading before reward points are cryptographically credited.

Q3: What happens if the university Wi-Fi or backend database goes down?

EcoLoop is architected with a Resilient Store pattern. If the primary cloud database is unreachable, the system gracefully operates on a synchronized local memory store with automatic retry queues. Furthermore, Next.js prerenders static assets and UI shells, ensuring the UI remains responsive and operational even in degraded connectivity.

Q4: How did you compute the CO2e emission savings and carbon certificates?

We implement the official Global E-waste Monitor (GEM 2024) Scope-3 methodology, which establishes an empirical emission avoidance factor of 1.44 kg of CO2 equivalent per 1.0 kg of certified recycled mixed electronics. The backend computes this synchronously upon recycler certificate generation, converting metric tons avoided into recognizable civic equivalents (car kilometers avoided and tree seedling offsets).

Q5: How does EcoLoop integrate with India's official CPCB EPR framework?

EcoLoop does not compete with or duplicate the central government portal; it acts as the digital first-mile ingestion conduit. By generating serialized custody manifests, university authorities receive verified receipts required to prove institutional compliance under India E-Waste Management Rules 2022 and NAAC Criterion 7.1.3.

Q6: How does your system scale across multiple universities or campuses?

The backend uses multi-tenant data segmentation. Each campus operates with isolated hostel models, student identifiers, and designated collection squads. Cross-campus leaderboards and regional municipal dashboards aggregate anonymized telemetry via REST endpoints, allowing Nitte to scale the platform across Bengaluru or nationwide.

Q7: Is user data safe under current Indian privacy laws?

Yes. EcoLoop strictly adheres to the Digital Personal Data Protection (DPDP) Act 2023. Passwords are salted and hashed using bcrypt (10 rounds), session tokens use signed JWTs with expiration bounds, and student identities are scrubbed from public leaderboards and municipal reporting feeds.

LIVE DEPLOYMENTS & OPEN-SOURCE REPOSITORY

- Live Web App: https://ecoloop-campus.vercel.app
- Live REST API: https://ecoloop-api-campus.vercel.app/api
- GitHub: https://github.com/ibaibhav0/ecoloop
- Swagger API Docs: https://ecoloop-api-campus.vercel.app/docs
- AI Audit Engine: /report
- CPCB Compliance Engine: /compliance